

Vitor Farias Costa de Carvalho

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Education

Purdue University

Master of Science in Industrial Engineering (with thesis), GPA: 3.86/4.0

08/2023 – 05/2025

Bachelor of Science in Industrial Engineering, minor: Mathematics, GPA: 3.74/4.0

08/2019 – 05/2023

Universidade Federal de Pernambuco

Bachelor of Science in Computer Engineering (1 semester completed)

01/2019 – 05/2019

Skills

Programming/Tools: Python, SQL, Power BI, Databricks, R, Microsoft Office, Excel, Git

Languages: English (full professional fluency), Portuguese (native), Spanish (proficient reading/writing; conversational)

Work Experience

Lafayette Urban Ministry (LUM)

Lafayette, IN

Data Analytics Intern

10/2025 – 12/2025

- Built a data pipeline in Python (Pandas) to convert WordPress form submissions into standardized CSV files for Bloomerang imports, automating onboarding of existing volunteers into the CRM database
- Improved Bloomerang database integrity by identifying, correcting, and merging inconsistent volunteer profiles
- Developed a step-by-step instructional guide to migrate volunteer opportunities from LUM's website to the Bloomerang Volunteer platform, streamlining hour tracking, volunteer check-in/out, and administrative workflow

Purdue University

West Lafayette, IN

Data Science Graduate Researcher (CONNplexity Lab)

08/2023 – 05/2025

- Developed a novel data-driven framework to identify individuals from brain imaging data, similar to a biological fingerprint
- Validated the approach on dataset with 426 participants, achieving accuracy levels that outperform previously published methods
- Led the project end-to-end, from methodology design and analysis to interpretation and manuscript preparation
- Published scientific article as the first author and presented findings at international conferences

Data Science Graduate Researcher (Programmable Structures Lab)

08/2023 – 05/2025

- Developed an object classification model using tactile data collected from sensor-embedded substrates
- Designed a framework to optimize sensor selection, achieving 90% classification accuracy while reducing sensor usage by 60%
- Applied signal-processing techniques to reconstruct object shapes from raw tactile measurements

Programming Lab Instructor

08/2023 – 12/2023

- Taught weekly lab sessions to 25 undergraduate students, covering fundamentals of SQL, R, HTML, CSS, JavaScript, and PHP
- Led live coding demonstrations, assisted students with lab/homework inquiries, and graded programming projects

Centro de Desenvolvimento de Sistemas

Brasília, DF

Software Engineering Intern

05/2023 – 08/2023

- Optimized the front-end of a Django application by implementing tables, menus, filters, and pagination, improving client-side rendering performance and reducing page load times
- Enhanced back-end architecture by improving views and URL routing, streamlining database interactions through ORM, and implementing robust form and error handling processes
- Refactored legacy code using Git, fixing 150+ bugs and improving code quality and documentation

Projects

Song Recommender System (Python, SQL, Databricks)

- Worked with a dataset of over 1.2 million Spotify songs to build a hybrid content-based recommendation system that suggests new tracks based on the audio characteristics of songs in a user's Spotify playlist
- Built a scalable data pipeline in Databricks to clean, structure, and enrich songs' metadata with Last.fm API popularity metrics, improving recommendation relevance

Personal Finance Dashboard (Power BI, Excel, Power Automate)

- Implemented an automated pipeline using Power Automate to process incoming credit card transaction emails, extract transaction date, merchant, and amount, and load them into a centralized Excel dataset
- Built a merchant classification system in Excel leveraging a custom mapping table to automatically assign spending categories (Food & Drink, Groceries, Shopping, etc.) to each transaction
- Developed an interactive mobile Power BI dashboard using DAX measures and automated daily refreshes to monitor spending patterns by category and merchant

PepsiCo Fleet MPG Optimization (Python)

- Conducted regression analyses in Python (Pandas, NumPy, Scikit-learn) in collaboration with colleagues, identifying key variables influencing fleet fuel efficiency
- Translated analytical results into business recommendations for optimizing fleet allocation, driving an estimated 20% reduction in operational costs while directly supporting PepsiCo's sustainability goals

COVID-19 Web Platform (R, SQL, Machine Learning)

- Collaborated on the development of a web application for respiratory disease diagnosis, featuring user authentication, MySQL based data management, symptom-driven prediction, and user feedback integration
- Applied machine learning techniques with cross-validation to evaluate model performance and used statistical resampling methods to address class imbalance